

Assignment 6b

Friday, 19 April 2024 6:34 pm

Assignment 6b

See Lecture 25

Q1, Q2, Q3.

Instead of Question 4, Do Exercise set 7.1 Q11, Q12

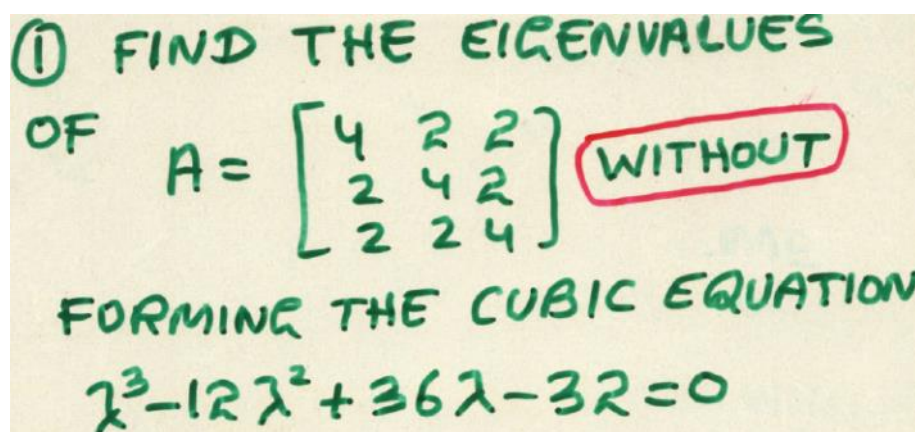
Instead of Question 5, do the Proof of Theorem 7.2.1 from our book

Q6, Q7, Q8

Instead of Q9, Do Exercise set 7.3, Q1

The rest of these questions are for Linear Transformations and will only be done after we have done the next two lectures.

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Solution: Considering linearly dependent equation that holds for eigenvalues in R^3 is

$$k_1 v + k_2 A v + k_3 A^2 v + k_4 A^3 v = 0 \quad \forall k_i \neq 0$$

Now let $v = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, so $A v = \begin{bmatrix} 4 \\ 2 \\ 2 \end{bmatrix}$, $A^2 v = \begin{bmatrix} 24 \\ 20 \\ 20 \end{bmatrix}$ and $A^3 v = \begin{bmatrix} 176 \\ 168 \\ 168 \end{bmatrix}$. So the matrix would be in the form of $k_1 v + k_2 A v + k_3 A^2 v + k_4 A^3 v = 0 \Rightarrow$

$$\begin{bmatrix} 1 & 4 & 24 & 176 \\ 0 & 2 & 20 & 168 \\ 0 & 2 & 24 & 176 \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Echelon form of above matrix leads to values for k_i .

$$\begin{aligned} k_1 &= 16k_3 + 160k_4 \\ k_2 &= -10k_3 - 84k_4 \\ k_3 &= t \\ k_4 &= s \end{aligned}$$

Lets take $s = 1$ and $t = -12$ for simplification of k_i . Hence $k_1 = 32, k_2 = 36, k_3 = -12$ and $k_4 = 1$.

Now substitute in above equation and solve for the roots of equation, we have 8,2,2.

② (SIMILAR MATRICES)

(i) IF A IS SIMILAR TO MATRIX B (A & B ARE SQUARE MATRICES) THEN B IS ALSO SIMILAR TO A .

(ii) SIMILAR MATRICES HAVE THE SAME DETERMINANT (PROVE IT) i.e.,

IF A IS SIMILAR TO B THEN $\det(A) = \det(B)$

DEFINITION: IF A AND B ARE SQUARE MATRICES, WE SAY B IS SIMILAR TO A IF THERE IS AN INVERTIBLE MATRIX P SUCH THAT $B = P^{-1}AP$.

$$\rightarrow A = \hat{Q} B Q$$

$$B = P^{-1} A P \Rightarrow P B P^{-1} = A$$

$$A = P B P^{-1} \Rightarrow \text{Let } P^{-1} = Q \text{ so, } Q^{-1} = P$$

$$A = Q^{-1} B Q$$

$$\begin{aligned} \det(B) &= \det(P^{-1} A P) \\ &= \det(P^{-1}) \det(A) \det(P) \\ &= \frac{1}{\cancel{\det(P)}} \det(A) \cancel{\det(P)} \end{aligned}$$

$$\det(B) = \det(A)$$

③ FIND A MATRIX P THAT DIAGONALIZES A ,

AND DETERMINE $P^{-1}AP$, WHERE

③ FIND A MATRIX P
THAT DIAGONALIZES A ,
AND DETERMINE $P^{-1}AP$, WHERE
 $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ ⑨

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$$\det(A - \lambda I) = 0 \Rightarrow (1-\lambda)((1-\lambda)(1-\lambda) - 1) = 0$$

$$\lambda = 1, \quad \lambda^2 - 2\lambda = 0 \Rightarrow \lambda = 2, 0$$

At $\lambda = 1$

$$\left[\begin{array}{ccc|ccc} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \end{array} \right] \Rightarrow \begin{array}{l} x_1 = x_1 \\ x_3 = 0 \\ x_2 = 0 \end{array} \Rightarrow \underline{X} = x_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

At $\lambda = 0$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

$$\rightarrow \begin{array}{l} x_1 = 0 \\ x_2 = -x_3 \\ x_3 = x_3 \end{array} \rightarrow \underline{X} = x_3 \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$$

At $\lambda = 2$

$$\left[\begin{array}{ccc|ccc} -1 & 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & 1 & 0 & 0 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} -1 & 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

$-v = 0$ $\{0\}$

$$\begin{aligned}
 -x_1 &= 0 \\
 x_2 &= x_3 \\
 x_3 &= x_3
 \end{aligned}
 \rightarrow x_3 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

Hence it is from symmetric Matrix & each vector is from different Eigenspace so, they are orthogonal

Just normalize them $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}, \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1/\sqrt{2} & 1/\sqrt{2} \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \text{ this is also symmetric } A^T = A = A^{-1}$$

$$A = P D P^{-1} \Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1/\sqrt{2} & 1/\sqrt{2} \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1/\sqrt{2} & 1/\sqrt{2} \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

11. Find the eigenvalues of A^9 for

$$A = \begin{bmatrix} 1 & 3 & 7 & 11 \\ 0 & \frac{1}{2} & 3 & 8 \\ 0 & 0 & 0 & 4 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

12. Find the eigenvalues and bases for the eigenspaces of A^{25} for

$$A = \begin{bmatrix} -1 & -2 & -2 \\ 1 & 2 & 1 \\ -1 & -1 & 0 \end{bmatrix}$$

For A

$$\begin{bmatrix} 2 & 1 \end{bmatrix}$$

For A^{25}

$$\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

for A

Eigenvalue: -1 , eigenvector: $\begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$.

Eigenvalue: 1 , eigenvector: $\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$.

Eigenvalue: 1 , eigenvector: $\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$.

for K

$$(-1)^{25} = -1$$

$$1^{25} = 1 \quad \text{same}$$

$$1^{25} = 1$$

In Exercises 3–7 use the method of Exercise 2 to determine whether the matrix is diagonalizable.

6. $\begin{bmatrix} -1 & 0 & 1 \\ -1 & 3 & 0 \\ -4 & 13 & -1 \end{bmatrix}$

7. $\begin{bmatrix} 2 & -1 & 0 & 1 \\ 0 & 2 & 1 & -1 \\ 0 & 0 & 3 & 2 \\ 0 & 0 & 0 & 3 \end{bmatrix}$

In Exercises 8–11 find a matrix P that diagonalizes A , and determine $P^{-1}AP$.

8. $A = \begin{bmatrix} -14 & 12 \\ -20 & 17 \end{bmatrix}$

Q6

Eigenvalue: 2 , eigenvector

Eigenvalue: $-\frac{i(\sqrt{3}-i)}{2}$,

Eigenvalue: $\frac{i(\sqrt{3}+i)}{2}$, e_i

Three different eigenvalues so three eigenvectors for sure, so it is diagonalizable

Q7

Eigenvalue: 3, eigenvector: $\begin{bmatrix} -1 \\ 1 \\ 1 \\ 0 \end{bmatrix}$.

Eigenvalue: 2, eigenvector: $\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$.

Repeated eigenvalues 3,3,2,2 and only two eigenvectors, so It is not diagonalizable.
(since the dimension of a matrix is not equal to the number of eigenvectors)

Q8

$$P = \begin{bmatrix} \frac{3}{4} & \frac{4}{5} \\ 1 & 1 \end{bmatrix}$$

$$D = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

Find the characteristic equation of the given symmetric matrix, and then by inspection determine the dimensions of the
1. eigenspaces.

(a) $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & -4 & 2 \\ -4 & 1 & -2 \\ 2 & -2 & -2 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

(d) $\begin{bmatrix} 4 & 2 & 2 \\ 2 & 4 & 2 \\ 2 & 2 & 4 \end{bmatrix}$

(e) $\begin{bmatrix} 4 & 4 & 0 & 0 \\ 4 & 4 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

(f) $\begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & 0 & 0 \\ 0 & 0 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix}$

1. (a) The characteristic equation is $\lambda(\lambda - 5) = 0$. Thus each eigenvalue is repeated once and hence each eigenspace is 1-dimensional.
- (c) The characteristic equation is $\lambda^2(\lambda - 3) = 0$. Thus the eigenspace corresponding to $\lambda = 0$ is 2-dimensional and that corresponding to $\lambda = 3$ is 1-dimensional.
- (e) The characteristic equation is $\lambda^3(\lambda - 8) = 0$. Thus the eigenspace corresponding to $\lambda = 0$ is 3-dimensional and that corresponding to $\lambda = 8$ is 1-dimensional.